

Milestone 1 Proposal: Identifying Maternal Health Access and Risk Patterns Across U.S. Counties

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DSC680 - Applied Data Science

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June 14, 2026

Topic

Identifying Maternal Health Access and Risk Patterns Across U.S. Counties. This project will analyze county-level maternal and birth-related indicators to explore where access gaps and elevated risk factors may point to areas that need stronger maternal health outreach and resource planning.

Business Problem

Maternal health outcomes in the United States continue to vary by geography, demographics, and access to care. Health systems, public health agencies, and community partners need clearer evidence about where maternal risk may be elevated, and which county-level factors are most associated with poor birth-related outcomes. This project will explore questions such as: Which county characteristics appear most associated with adverse maternal or infant-related indicators? Are there patterns in prenatal care, maternal age, rurality, workforce availability, or socioeconomic conditions that align with worse outcomes? How can those patterns support more targeted outreach, planning, and intervention?

Datasets

The primary dataset will be CDC WONDER Natality data because it provides county-level U.S. birth and maternal variables derived from birth certificates, including maternal age, race, prenatal care timing, gestation, tobacco use, delivery method, birth weight, and geography. County-level context will be added from the HRSA Area Health Resources Files, which include provider supply, facilities, population characteristics, economics, and other health resource measures. If needed, HRSA shortage designation data, such as Primary Care HPSA or Maternity Care Target Area indicators, may be added to represent maternity care access constraints better.

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This combination is a stronger fit for county-level analysis than a broader composite source because it directly links birth outcomes with local health resource conditions.

Methods

The project will begin with data cleaning, variable selection, and dataset merging at the county level using common geographic identifiers. Exploratory data analysis will be used to identify distributions, outliers, regional patterns, and missingness. Correlation analysis and comparative visualizations will be used to examine relationships between birth-related outcome indicators and potential drivers, including access to care, demographics, and local resource availability. Depending on data quality and scope, the project may also use regression or classification methods to estimate which features are most associated with higher-risk counties. Final outputs will likely include summary tables, maps, and explanatory visualizations designed for a nontechnical audience.

Ethical Considerations

Maternal health analysis carries important ethical concerns. Geographic and demographic patterns can be oversimplified or framed in ways that stigmatize communities. Publicly available data may still mask structural causes if the analysis focuses too much on individual-level factors. There is also a risk of treating correlation as causation, especially when county-level data are aggregated and cannot explain each patient's experience. To reduce these risks, the project will use de-identified public data, avoid spectacular claims, clearly label limitations, and frame disparities as system-level and access-related issues rather than personal failures.

Challenges & Issues

A likely challenge will be aligning datasets collected at different times, with different file structures and definitions. County-level health data may contain suppressed values, missing fields, or inconsistent variable names across sources. HRSA shortage designations may not align perfectly with county boundaries in every case, which may require careful aggregation or the use of broader county-level access proxies from AHRF. Another challenge will be balancing a meaningful model with careful interpretation, since many overlapping structural factors often influence public health relationships.

References

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